

Hyeonjun An

Ph.D. Candidate in Physics | Ulsan National Institute of Science and Technology

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RESEARCH PROFILE

I use quantitative neutron and synchrotron X-ray imaging to resolve mass transport and phase changes inside soft materials. My research focuses on polymer swelling and drying, water-vapor uptake by hygroscopic liquids, freezing of polymer and saline droplets, and liquid imbibition in heterogeneous porous materials.

EDUCATION

Combined M.S. and Ph.D. in Physics

Feb 2022 – Present

Ulsan National Institute of Science and Technology (UNIST)

Expected graduation: February 2027 | Advisor: Prof. Joonwoo Jeong | GPA: 3.93/4.3

B.S. in Physics and Energy Engineering

Mar 2016 – Feb 2022

UNIST | Double major

Summa cum laude | GPA: 3.91/4.3

Student Exchange Program

Summer 2017

Hong Kong University of Science and Technology (HKUST), Hong Kong

RESEARCH EXPERIENCE

Doctoral Researcher

Feb 2022 – Present

Soft Matter Physics Laboratory, UNIST | Advisor: Prof. Joonwoo Jeong

- Polymer swelling and drying. Measure solvent uptake in hydrogels and PDMS using neutron isotope contrast and inverse Abel reconstruction; compare internal composition fields with poromechanical transport models.
- Hygroscopic liquids. Quantify water-vapor uptake and concentration gradients in droplets and capillaries; examine gravity, diffusion and mixing.
- Droplet freezing. Track freezing fronts and reconstruct frozen structures by synchrotron radiography and tomography; study polymer redistribution, brine networks, bubbles and apex morphology.
- Porous materials. Quantify the evolution of liquid distributions and wetting fronts during imbibition of evaporative liquids in heterogeneous media.

Research Placement, Pohang Accelerator Laboratory

Apr – Dec 2024

Synchrotron Utilization Research Project | PLS-II 6C BMI; separate 50-day research placement.

Research Intern, KRISS

Jan – Feb 2022

Multiscale Length Metrology Team | Femtosecond-laser-based precision length measurement | Supervisor: Dr. Jonghan Jin

Research Intern, UNIST

Mar – Dec 2021

Soft Matter Physics Laboratory | Microfluidics research | Supervisor: Prof. Joonwoo Jeong

PUBLICATIONS AND MANUSCRIPTS

Published articles and conference proceedings

Hyosim Yang, Gangseon Ji, Min Choi, Seondo Park, [Hyeonjun An](#), Hyoung-Taek Lee, Joonwoo Jeong, Yun Daniel Park, Kyungwan Kim, Noejung Park, Jeeyoon Jeong, Dai-Sik Kim, Hyeong-Ryeol Park.

Suppressed terahertz dynamics of water confined in nanometer gaps.

Science Advances 10, eadm7315 (2024).

doi:10.1126/sciadv.adm7315

[Hyeonjun An](#), Youngtak Koo, Jiyong Cheon, Pavel Trtik, Joonwoo Jeong.

Condensation of the Vapor in Sessile Water Droplets and Open-Ended Water-Filled Capillaries Revealed by Neutron Radiography.

Proceedings of the 12th World Conference on Neutron Radiography. Springer Proceedings in Physics 348, 235–243 (2026).

doi:10.1007/978-3-032-15003-5_27

Manuscripts under review

[Hyeonjun An](#), Bomi Kim, Jae Kwan Im, Sungwoo Kim, Min Woo Kim, Seob-Gu Kim, Jae-Hong Lim, Kitae Kim, Joonwoo Jeong.

Polymer-Regulated Freezing of Water Droplets Revealed by Synchrotron X-ray Imaging and Raman Spectroscopy.

Manuscript under review.

Earlier preprint: arXiv:2606.01251.

Author list shown for the revised manuscript; the earlier preprint has a different author list.

Jae Kwan Im, [Hyeonjun An](#), Seob-Gu Kim, Jae-Hong Lim, Joonwoo Jeong.

X-ray Microscopy Study of Freezing Sessile Droplets.

Manuscript under review.

Preprint: arXiv:2511.20571.

NEUTRON AND SYNCHROTRON EXPERIENCE

Neutron imaging

70 beamtime days

PSI ICON and NEUTRA, Switzerland; HANARO ENF, Korea.

Quantitative radiography, H/D isotope contrast, attenuation-based composition analysis and experimental coordination.

Synchrotron X-ray imaging

15 beamtime days

PAL PLS-II 6C BMI, 9D and 7C, Korea.

Time-resolved radiography and tomography. Separate 50-day research placement at PAL 6C; not included in the beamtime total.

Synchrotron scattering and diffraction

PAL PLS-II 6D and 9A, Korea.

TECHNICAL SKILLS

- Imaging and analysis: neutron attenuation, H/D contrast, inverse Abel reconstruction, X-ray tomography, segmentation and three-dimensional structural characterization.
- Experiments: temperature-controlled droplet experiments, freezing-front tracking, microfluidics, optical and Raman characterization.
- Modeling: Flory–Rehner and Darcy descriptions of swelling; multicomponent diffusion and Maxwell–Stefan descriptions of water uptake.
- Workflow: beamtime proposals, international coordination, quantitative data analysis and LaTeX manuscript preparation.

RESEARCH FUNDING

Principal Investigator

Aug 2026 – Feb 2027

2026 | Joint Research Program for Ph.D. Students Utilizing Overseas Research Facilities

Korea Nuclear International Cooperation Foundation (KONICOF) | KRW 30,000,000

Project: Quantitative neutron imaging of mass transport in soft matter with complementary X-ray imaging.

International partner: Paul Scherrer Institute, Switzerland. Contract period: 18 August 2026 – 17 February 2027.

PARTICIPATING RESEARCH PROJECTS

2023–2024 | National Research Foundation of Korea (NRF) – Mid-Career Researcher Program

Active self-organization of living matter under confinement with curvatures.

2023–2024 | National Research Foundation of Korea (NRF) – Mid-Career Researcher Program

Advanced study and application of water-based anisotropic medium: Phase behavior and control of nonequilibrium phenomena.

2024 | Pohang Accelerator Laboratory (PAL) – Synchrotron Radiation Utilization Research Program

X-ray imaging study of nonequilibrium soft matter: Phase transitions of water.

2024–2026 | National Research Foundation of Korea (NRF) – Excellent Research Program (Core Research, Type B)

Active bath: Characterization and spatiotemporal control of nonequilibrium properties.

2024–2026 | National Research Foundation of Korea (NRF) – Global Matching Program (Germany)

Exploring dilatancy in soft granular materials using advanced radiography techniques.

2024–2026 | Korea Polar Research Institute (KOPRI) – Academic-Institute Polar Research Promotion Program (PAP)

High-resolution radiography study of polar ice: Microstructure and phase transition.

AWARDS AND SCHOLARSHIPS

Excellence Award

Aug 2026

DRB Space Challenge · Drop-tower microgravity experiment

Misan Mountaineering Scholarship

Fall 2026

UNIST

Outstanding Presentation Award in Oral Session

Apr 2026

Korean Physical Society Spring Meeting · Polymer swelling

Outstanding Presentation Award in Oral Session

Apr 2025

Korean Physical Society Spring Meeting · Water uptake in hygroscopic liquids

UNIST Scholarship (Graduate)

2021–2026

UNIST

UNIST Scholarship (Teaching Assistant)

2021–2026

UNIST

Summa cum laude

Feb 2022

Bachelor of Science · UNIST

National Science and Engineering Undergraduate Scholarship

2016–2018; 2021

Korea Student Aid Foundation (KOSAF)

CONFERENCE CONTRIBUTIONS

Contributed oral presentations

Korean Physical Society Spring Meeting Apr 2026

“Quantifying Polymer Swelling Kinetics via Neutron Radiography.” Outstanding Presentation Award.

Korean Physical Society Spring Meeting Apr 2025

“Direct Observation of Water Vapor Uptake in Hygroscopic Liquids by Neutron Radiography.” Outstanding Presentation Award.

Poster presentations

International HANARO Symposium Sep 2025

“Water Vapor Uptake into Hygroscopic Liquids in Capillaries: Neutron Radiography Study.”

APCTP/G-LAMP Workshop on the Roles of Heterogeneity in Nonequilibrium (RHINO) Aug 2025

“How Evaporative Liquids Soak into Heterogeneous Porous Materials: A Neutron Radiography Study.”

Ice at the Micro-scale Jul 2025

“X-ray Microscopy Study of Freezing Sessile Droplets: Effects of Dissolved Polymer.”

12th World Conference on Neutron Radiography (WCNR-12) Jun 2024

“Direct Observation of Water Uptake by Hygroscopic Liquid Droplets with Neutron Radiography.”

Idaho Falls, USA. Presented by Pavel Trtik.

Korean Physical Society Spring Meeting Apr 2023

“Direct Observation of Water Uptake by Hygroscopic Liquid Droplets with Neutron Radiography.”

Upcoming poster presentations (as of 6 October 2026)

10th International Topical Meeting on Neutron Radiography (ITMNR-10) Oct 2026

“Quantifying Polymer Swelling Kinetics via Neutron Radiography.”

10th International Topical Meeting on Neutron Radiography (ITMNR-10) Oct 2026

“Direct Observation of Water Vapor Uptake in Hygroscopic Liquids by Neutron Radiography.”

Korea Physical Society Fall Meeting Oct 2026

“Connectivity of Brine Networks and Tip Morphology in Freezing Saline Droplets.”

10th International Topical Meeting on Neutron Radiography (ITMNR-10): two accepted posters; Annecy, France, 12–16 October 2026.

TEACHING EXPERIENCE

Teaching Assistant, UNIST

General Physics Laboratory I — Spring & Fall 2024; Spring & Fall 2025; Spring 2026.

Mathematical Physics — Fall 2023. Prof. Chae Un Kim.

Thermal and Statistical Physics — Spring 2023. Prof. Yoon Seok Oh.

General Physics I — Spring 2018; Spring & Fall 2022. Undergraduate teaching assistant in Spring 2018.

General Physics II — Fall 2018. Undergraduate teaching assistant.

MILITARY SERVICE

Republic of Korea Air Force 2019–2021

Motor transport operator, 8th Fighter Wing.